

Further operations with vectors — Problem Set

Warm-up

1. Calculate the dot product $\underline{u} \cdot \underline{v}$ where $\underline{u} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\underline{v} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$.
2. Find the value of k such that the vectors $\begin{pmatrix} k \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ -6 \end{pmatrix}$ are perpendicular.
3. Determine the magnitude of the vector $\underline{w} = 5\underline{i} - 12\underline{j}$.
4. Given $\underline{a} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$, calculate the scalar $\underline{a} \cdot \underline{b}$.
5. State the unit vector $\hat{v}$ in the direction of $\underline{v} = \begin{pmatrix} 8 \\ 6 \end{pmatrix}$.
6. If $|\underline{u}| = 4$, $|\underline{v}| = 7$ and the angle between them is 60° , find $\underline{u} \cdot \underline{v}$.
7. Determine if the vectors $\underline{p} = 2\underline{i} - 3\underline{j}$ and $\underline{q} = 6\underline{i} + 4\underline{j}$ are perpendicular.
8. Write down the formula for the vector projection of $\underline{a}$ onto $\underline{b}$.
9. Find the scalar component of $\underline{u} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$ in the direction of $\underline{i}$.
10. Simplify the expression $\underline{a} \cdot \underline{a}$ in terms of the magnitude of $\underline{a}$.

Skill-building

1. Let $\underline{u} = 2\underline{i} + \underline{j}$ and $\underline{v} = \underline{i} - 3\underline{j}$. Find the vector projection of $\underline{u}$ onto $\underline{v}$.
2. Find the acute angle between the vectors $\underline{a} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$ to the nearest degree.
3. The vectors $\begin{pmatrix} m+2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ m \end{pmatrix}$ are perpendicular. Find all possible values of m .
4. Given $A(1, 2)$, $B(4, 6)$, and $C(-3, 5)$, find the vector $\overrightarrow{AB}$ and calculate $\overrightarrow{AB} \cdot \overrightarrow{AC}$.
5. A force $\vec{F} = \begin{pmatrix} 3 \\ 7 \end{pmatrix}$ acts on an object moving it from $(0, 0)$ to $(2, -1)$. Calculate the work done $W = \vec{F} \cdot \vec{s}$.
6. Find a unit vector that is perpendicular to $\underline{v} = 5\underline{i} - 12\underline{j}$.

7. Show that for any non-zero vector $\underline{a}$, the vector $\underline{p} = \underline{u} - \text{proj}_{\underline{a}} \underline{u}$ is perpendicular to $\underline{a}$.
8. If $\underline{u} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$ and $\text{proj}_{\underline{v}} \underline{u} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, find a possible direction vector for $\underline{v}$.
9. Two vectors $\underline{a}$ and $\underline{b}$ are such that $|\underline{a}| = 3$, $|\underline{b}| = 5$, and $|\underline{a} + \underline{b}| = 7$. Find the angle between $\underline{a}$ and $\underline{b}$.
10. Find the vector projection of $\underline{i} + \underline{j}$ onto the line $y = 2x$.

Easier Exam Questions

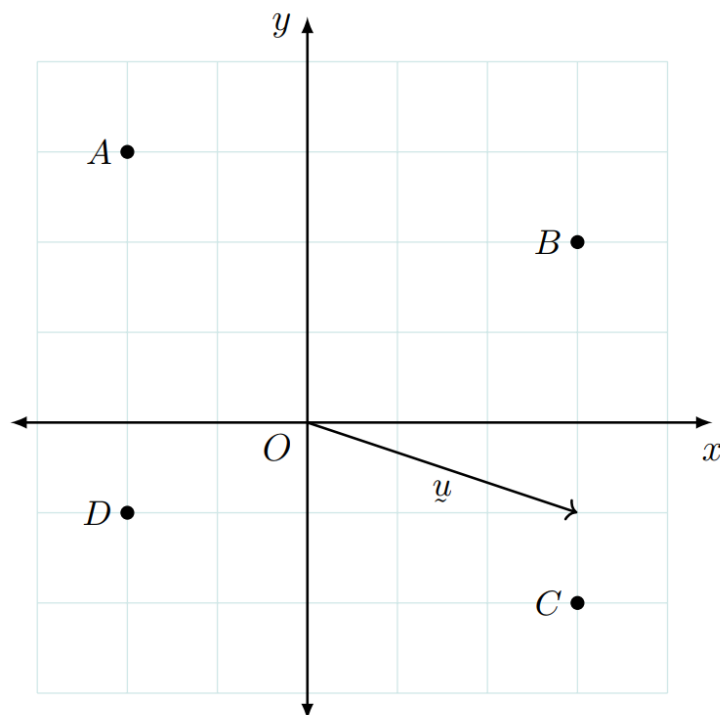
1. (Abbotsleigh 2024 Q2) Which of the following vectors are perpendicular?
 - (A) $\underline{u} = 3\underline{i} - 7\underline{j}$, $\underline{v} = \frac{7}{3}\underline{i} - \underline{j}$
 - (B) $\underline{u} = 14\underline{i} + \underline{j}$, $\underline{v} = \frac{1}{2}\underline{i} - 7\underline{j}$
 - (C) $\underline{u} = \frac{9}{2}\underline{i} + \frac{7}{2}\underline{j}$, $\underline{v} = -\frac{2}{9}\underline{i} - \frac{1}{7}\underline{j}$
 - (D) $\underline{u} = \frac{2}{5}\underline{i} - 2\underline{j}$, $\underline{v} = 5\underline{i} + 3\underline{j}$
2. (Girraween 2020 Q2) A vector perpendicular to $3\underline{i} + 4\underline{j}$ and with magnitude 5 is
 - (A) $-4\underline{i} - 3\underline{j}$
 - (B) $5(4\underline{i} - 3\underline{j})$
 - (C) $4\underline{i} - 3\underline{j}$
 - (D) $\sqrt{5}(4\underline{i} - 3\underline{j})$
3. (Baulkham Hills 2023 Q3) Which vector is perpendicular to $\underline{v} = 2\underline{i} - 5\underline{j}$?
 - (A) $\underline{a} = -5\underline{i} + 2\underline{j}$
 - (B) $\underline{b} = 5\underline{i} - 2\underline{j}$
 - (C) $\underline{c} = 2\underline{i} + 5\underline{j}$
 - (D) $\underline{d} = 5\underline{i} + 2\underline{j}$
4. (Newington 2024 Q4) What is the size of the angle between the vectors $\underline{a} = 3\underline{i} + \underline{j}$ and $\underline{b} = 2\underline{i} - \underline{j}$?
 - (A) $\frac{\pi}{6}$
 - (B) $\frac{\pi}{4}$
 - (C) $\frac{\pi}{3}$
 - (D) $\frac{\pi}{2}$

5. (Fort St 2024 Q2) The acute angle (to the nearest degree) between the vectors $\underline{u} = \begin{pmatrix} \sqrt{3} \\ 2 \end{pmatrix}$ and $\underline{v} = \begin{pmatrix} 9 \\ \sqrt{17} \end{pmatrix}$ is:
- (A) 24°
 (B) 51°
 (C) 46°
 (D) 48°
6. (Cheltenham Girls 2024 Q11a)
 Show that the vectors $\underline{s} = \begin{pmatrix} 5 \\ 10 \end{pmatrix}$ and $\underline{t} = \begin{pmatrix} -4 \\ 2 \end{pmatrix}$ are perpendicular. **2**
7. (Hurlstone 2021 Q11e) Find the vector projection of $\underline{a}$ onto $\underline{b}$ if $\underline{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\underline{b} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$. **2**
8. (Girraween 2023 Q3) Given the vectors $\underline{u} = \begin{pmatrix} -2 \\ -5 \end{pmatrix}$ and $\underline{v} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$, the vector projection of $\underline{u}$ on $\underline{v}$ is;
- (A) $-\frac{11}{9} \begin{pmatrix} 3 \\ 1 \end{pmatrix}$
 (B) $-\frac{11}{10} \begin{pmatrix} 3 \\ 1 \end{pmatrix}$
 (C) $-\frac{39}{10} \begin{pmatrix} 3 \\ 1 \end{pmatrix}$
 (D) $\frac{1}{10} \begin{pmatrix} 3 \\ 1 \end{pmatrix}$
9. (Hurlstone 2024 Q4) What is the vector projection of $\underline{a} = -3\underline{i} + 4\underline{j}$ onto $\underline{b} = 2\underline{i} - \underline{j}$?
- (A) $2\underline{i} - 4\underline{j}$
 (B) $4\underline{i} - 2\underline{j}$
 (C) $-2\underline{i} + \underline{j}$
 (D) $-4\underline{i} + 2\underline{j}$
10. (Blacktown Boys 2020 Q3) What is the angle between the vectors $\begin{pmatrix} -3 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 18 \end{pmatrix}$?
- (A) $\theta = \cos^{-1} \left(\frac{39}{65} \right)$
 (B) $\theta = \cos^{-1} \left(-\frac{39}{65} \right)$
 (C) $\theta = \cos^{-1} \left(\frac{33}{65} \right)$
 (D) $\theta = \cos^{-1} \left(-\frac{33}{65} \right)$

11. (Blacktown Boys 2022 Q7) What is the vector projection of $\mathbf{p} = \begin{pmatrix} 5 \\ -2 \end{pmatrix}$ onto $\mathbf{q} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$?
- (A) $\mathbf{i} + \mathbf{j}$
 (B) $\frac{56}{100}\mathbf{i} + \frac{42}{75}\mathbf{j}$
 (C) $\frac{56}{25}\mathbf{i} + \frac{42}{25}\mathbf{j}$
 (D) $56\mathbf{i} + 42\mathbf{j}$

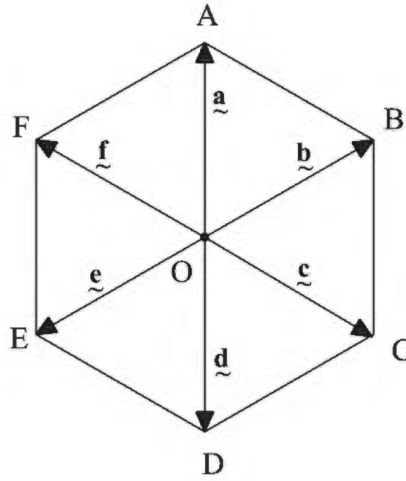
Harder Exam Questions

1. (Sydney Girls 2025 Q3) The vectors $\underline{\hat{a}}$ and $\underline{\hat{b}}$ satisfy $\underline{\hat{a}} \cdot \underline{\hat{b}} = -\frac{1}{2}$. Which of the following is the angle between $\underline{\hat{a}}$ and $\underline{\hat{b}}$?
- (A) $\frac{\pi}{3}$
 (B) $\frac{2\pi}{3}$
 (C) $\frac{5\pi}{6}$
 (D) $\frac{3\pi}{4}$
2. (Ascham 2024 Q5)
 Which of the following vectors has the greatest magnitude?



- (A) $\text{proj}_{\underline{\hat{u}}} \overrightarrow{OA}$
 (B) $\text{proj}_{\underline{\hat{u}}} \overrightarrow{OB}$
 (C) $\text{proj}_{\underline{\hat{u}}} \overrightarrow{OC}$
 (D) $\text{proj}_{\underline{\hat{u}}} \overrightarrow{OD}$

3. (Sydney Tech 2023 Q10) $ABCDEF$ is a regular hexagon made from 6 equilateral triangles with side-lengths of 2 units as shown below. $\vec{OA} = \mathbf{a}$, $\vec{OB} = \mathbf{b}$, $\vec{OC} = \mathbf{c}$, $\vec{OD} = \mathbf{d}$, $\vec{OE} = \mathbf{e}$, and $\vec{OF} = \mathbf{f}$



What is the value of $\text{proj}_{\mathbf{c}} \mathbf{a} \cdot \text{proj}_{\mathbf{f}} \mathbf{e}$ (the dot product of the vectors formed by projecting $\mathbf{a}$ and $\mathbf{e}$ onto $\mathbf{c}$ and $\mathbf{f}$ respectively)?

- (A) -2
(B) -1
(C) 1
(D) 2

4. (Hornsby Girls 2020 Q12c) The work done W , by a constant force $\vec{F}$, in moving a particle through a displacement $\vec{s}$, is defined by the formula $W = \vec{F} \cdot \vec{s}$. A force described by the vector $\vec{F} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$ moves a particle along the line l from $P(-1, 2)$ to $Q(2, -2)$ so that $\vec{s} = \overrightarrow{PQ}$.

- (i) Find $\vec{s}$ and hence find the value of W . 1
(ii) Hence, verify that W is also given by $W = (\vec{F} \cdot \hat{s})|\vec{s}|$. 1
(iii) Find the **vector projection** of $\vec{F}$ onto $\vec{s}$. 1

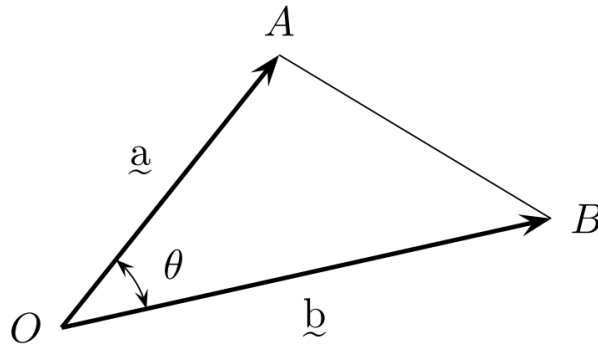
5. (Barker 2020 Q12a) Evaluate x if the vectors $\begin{pmatrix} x \\ 3 \end{pmatrix}$ and $\begin{pmatrix} -6 \\ 8 \end{pmatrix}$ are:

- (i) perpendicular 1
(ii) parallel 1

6. (Cheltenham 2023 Q8) Of the vectors $\underline{u} = (-2, 6)$, $\underline{v} = (9, 3)$ and $\underline{w} = (3, 2)$, which pairs are perpendicular?

- (A) Only $\underline{u}$ and $\underline{v}$ are perpendicular.
(B) Only $\underline{u}$ and $\underline{w}$ are perpendicular.
(C) Only $\underline{v}$ and $\underline{w}$ are perpendicular.
(D) There are no pairs of vectors which are perpendicular.

7. (Moriah 2023 Q10c) The vector $\underline{u}$ starts at the origin and ends at $3\underline{i} + 2\underline{j}$.
The vector $\underline{v}$ starts at $2\underline{j}$ and ends at $3\underline{i} + \lambda\underline{j}$.
- (i) Given that $\underline{u}$ is perpendicular to $\underline{v}$, show that $\lambda = \frac{-5}{2}$. **3**
- (ii) Hence find the length of $|\underline{v}|$. **2**
8. (Normanhurst Boys 2020 Q12b) A force, described by the vector $F_{\sim} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$, moves a particle along the line ℓ from $A(-1, -1)$ to $B(2, -3)$.
- (i) Find the unit vector that is in the direction of $\overrightarrow{AB}$. **2**
- (ii) Find the component of $F_{\sim}$ in the direction of line ℓ . **2**
9. (Normanhurst Boys 2020 Q12c)
- The diagram below shows $\triangle OAB$, where $\overrightarrow{OA} = \underline{a}$ and $\overrightarrow{OB} = \underline{b}$. **3**



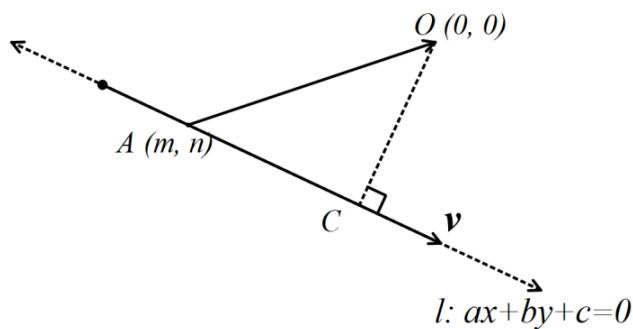
Given that θ is the angle between vectors $\underline{a}$ and $\underline{b}$, show that the area of the triangle is

$$A = \frac{1}{2} \sqrt{(\underline{a} \cdot \underline{a})(\underline{b} \cdot \underline{b}) - (\underline{a} \cdot \underline{b})^2}$$

10. (Blacktown Boys 2024 Q10) Given that $\underline{a}$ and $\underline{b}$ are two non-zero vectors, let $\underline{c}$ be the projection of $\underline{a}$ onto $\underline{b}$.
What is the projection of $12\underline{a}$ onto $3\underline{b}$?
- (A) $36\underline{c}$
(B) $12\underline{c}$
(C) $4\underline{c}$
(D) $3\underline{c}$
11. (Moriah 2022 Q11b) The vectors $\underline{u} = \begin{pmatrix} 6 \\ a \end{pmatrix}$ and $\underline{v} = \begin{pmatrix} a+1 \\ -9 \end{pmatrix}$ are perpendicular. **2**
- Find the value of a .
12. (Baulkham Hills 2020 Q9) If $\underline{u}$ and $\underline{v}$ are two non-zero vectors on the Cartesian plane, which of the following statements is **NOT ALWAYS CORRECT** ?

- (A) $|\text{proj}_{\tilde{v}} \tilde{u}| \leq |\tilde{u}|$
 (B) $|\text{proj}_{\tilde{v}} \tilde{u}| \leq |\tilde{v}|$
 (C) $|\text{proj}_{\tilde{v}} \tilde{u}| = \frac{|\tilde{u} \cdot \tilde{v}|}{|\tilde{v}|}$
 (D) $|\text{proj}_{\tilde{v}} \tilde{u}| = |\tilde{u} \cdot \hat{\tilde{v}}|$

13. (Hornsby Girls 2023 Q14d) Point $A(m, n)$ is on the line $l : ax + by + c = 0$. The values of m, n, a, b, c are all real constants.
 Vector $\tilde{v}$ overlaps line l .



- (i) Show that $\tilde{v} = \begin{pmatrix} -b \\ a \end{pmatrix}$ is a vector that overlaps line l . **1**
 (ii) Using vector projection, show that the perpendicular distance from O to line l is **4**

$$\frac{|c|}{\sqrt{a^2 + b^2}}$$